SYNTHESIS GAS

CPT : Petroleum products, petrochemicals, hydrocarbons, gasoline, petrol etc.

First generation petrochemicals → obtained directly from petroleum.

① synthesis gas $[CO + H_2]$ is a raw material for 1st generation petrochemicals. It is obtained from steam reforming of hydrocarbons

② Olefins (also raw material for 1st generation) are obtained from Pyrolysis of hydrocarbons.

[RAW MATERIALS for 1st gen. petrochemicals]

③ Aromatics from catalytic reforming of gasoline.

It is difficult to obtain pure $CO + H_2$; some CO_2 is always present. [which is sometimes beneficial too].

H_2 to $(CO + CO_2)$ ratio is to be analysed

$\downarrow$

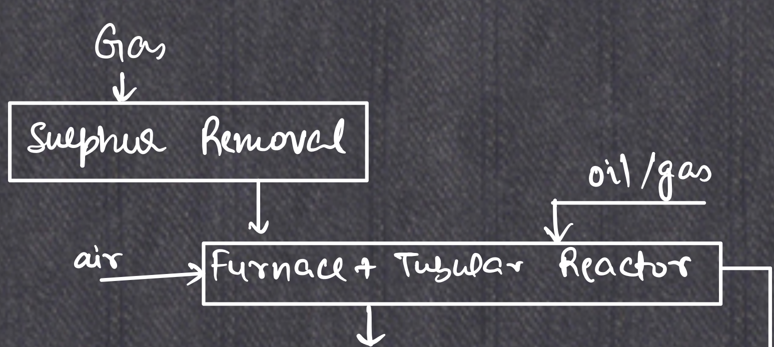
most precious product.

Steam reforming of methane : (SRM)

$$CH_4 + H_2O \rightleftharpoons CO + H_2 \qquad \Delta H = +49 \text{ kcal/mol}$$

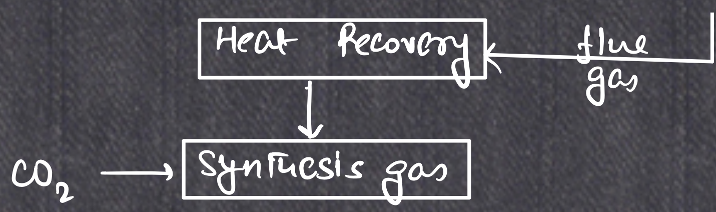
water gas shift reaction / shift conversion :

$$CO + H_2O \rightleftharpoons CO_2 + H_2$$



Composition of synthesis gas from methane steam reforming :-

Components	vol %
CO	15.5
CO_2	8.1
H_2	[illegible]



..2	75.7
N_2	0.2
CH_4	0.5
H_2S	~0

catalyst:

Ni / Al_2O_3 / Ca-aluminate / Mg-aluminate

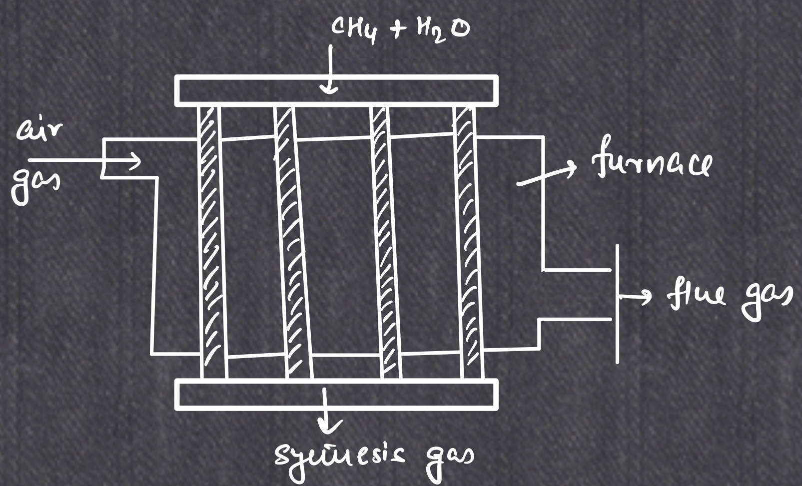
(6-20% by wt.)

HEAT RECOVERY

Product gas exchanges heat with feedstock. High pressure steam used in SRM etc.

Shift reaction: $CO + H_2O \longrightarrow H_2 + CO_2$ (Iron-chromate catalyst)

Tubular reactor for steam reforming



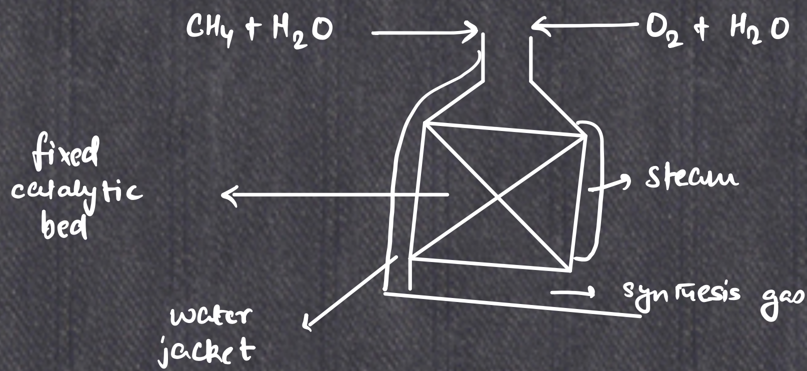
Sometimes, heat adjustment is done by using a part of the feed itself as fuel. (by burning it).
[Autothermal reaction]

Fuel gas : oxygen = 1 : 0.55

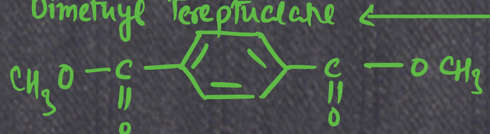
In an oxidative/autothermal process, furnace is usually not required.
Hydrocarbon : O_2 ≈ 1 : 0.55

A part of HC is burnt in the reactor to make-up the heat loss by the endothermic S.R. reaction by the endothermic combustion of HC.

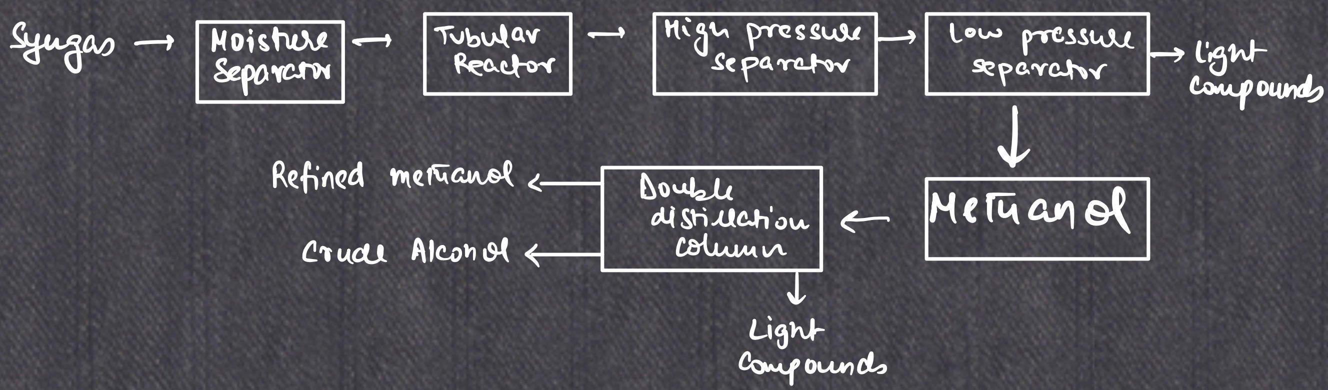
Shaft - Continuous Reactor :



Methane →
- Hydrocarbon Cyanide (HCN)
- Carbon Disulphide (CS_2) →
  - Hydrogen
  - Phosgene [$COCl_2$]
  - Ethylene Glycol (CH_2OH | CH_2OH)
  - Oxoalcohols (R–OH)
  - Methanol (CH_3OH) →
    - Ammonia (NH_3) →
      - Urea
      - Nitric Acid
      - Hydrazine
      - Methyl Amines
      - Alkanol Amines
      - Hexamethylenetetramine (Urotropin)
      - Acrylonitrile
    - Formaldehyde (HCHO)
    - Methyl chloride (CH_3Cl)
    - Methyl Amine (CH_3NH_2)
    - Methyl methacrylate: $H_2C=C(CH_3)-C(=O)-OCH_3$
    - Dimethyl Terephthalate

      
    - Acetic Acid CH_3COOH
- Synthesis Gas ($CO + H_2$)
- Chlorinated Methane (CH_3Cl, CH_2Cl_2, $CHCl_3$, CCl_4)
- Single cell protein.

Methanol



Reactions:

Exothermic
$$\begin{cases} CO + 2H_2 \rightleftharpoons CH_3OH & \Delta H = -90.8 \text{ kJ/mol} \\ CO_2 + 3H_2 \rightleftharpoons CH_3OH + H_2O & \Delta H = -49.5 \text{ kJ/mol} \end{cases}$$

$$CO_2 + H_2 \rightleftharpoons CO + H_2O \quad \text{(side reaction)}$$

Methanation

$$CO + 3H_2 \rightleftharpoons CH_4 + H_2O$$

Methyl ether formation

$$2CH_3OH \rightleftharpoons CH_3OCH_3 + H_2O$$

SIDE REACTIONS TO BE AVOIDED.

Carbon Efficiency / Selectivity of methanol

$$\frac{\text{No. of } CH_3OH \text{ moles produced}}{\text{No. of } (CO + CO_2) \text{ moles in syngas}} \times 100\%$$

Production of methanol is favoured by :-

- → elevation of pressure.
- → Reduction of temperature.
- → Increase of CO/CO_2 ratio.
- → Increase of H_2 content in syngas.

Thorium + copper $\xrightarrow{\text{molten}}$ $ThCu_6$ → New Catalyst merged

(Intermetallic)

Continuous decrease in undesired side reactions, and production of methanol. / 10 times more active than Cu.

Very good stability. Upto 475 hrs sustains it's activity.

Ethanol

Uses

→ Fuel Additive (additive to petroleum fuel, gasoline etc. to raise vapour pressure in cold environment)

→ As a solvent in the laboratory/industry.

→ Octane improvement

Diesel is used for compression/ignition engine. Octane number for fuels. Ethanol has very high octane no.

Formation

→ from agricultural waste by fermentation of carbohydrates

→ from ethylene [indirect hydration
direct hydration

Ethanol production -

① Fermentation process from agricultural waste.

② Synthesis from ethylene:

(a) Indirect hydration by H_2SO_4.

(b) direct hydration by steam.

$H_2C=CH_2 + \underset{(conc.)}{H_2SO_4} \xrightarrow{exothermic} C_2H_5O-SO_3H$ (Ethyl acid sulphate)

$$2\ H_2C=CH_2 + \underset{(conc.)}{H_2SO_4} \longrightarrow (C_2H_5O)_2SO_4 \quad \text{(diethyl sulphate)}$$

The ethyl / ethyl acid sulphates are hydrolysed.

Hydrolysis:

$$C_2H_5OSO_3H \xrightarrow{H_2O} C_2H_5OH + H_2SO_4$$

$$(C_2H_5O)_2SO_4 \xrightarrow{2H_2O} 2C_2H_5OH + H_2SO_4$$

$+ (C_2H_5)_2O$ small amount diethyl ether (5-10 %)

Re-conc. of dil. H_2SO_4:

Evaporation under vacuum, to get ~98% conc. H_2SO_4.

Feed

35 vol % C_2H_4 + inert gases like CH_4, C_2H_6

↓ counter-current through absorber

~80°C, 13-15 atm

inside 98% conc. H_2SO_4

(Above 90°C, resins and oils are formed. So temp. should be controlled. Reactor should have agitation to ensure completion).

Absorber is cooled externally. Proper control on temp. and exact amount of process water.

Problems:

Corrosion due to conc. H_2SO_4, production of Et-O-Et, high maintenance.

↓

mixture of Et_2O and EtOH is undesired.

Direct hydration:

$$C_2H_4(g) + H_2O(g) \longrightarrow C_2H_5OH(g) + \Delta H$$

$\Delta H = -43.4$ kJ/mol

Side reaction of Et-O-Et at low temp.] → $C_2H_5OH + C_2H_4 \longrightarrow Et\text{-}O\text{-}Et$

- By LCP, increasing pressure should favour the reaction but at high pressure, EtOH polymerises (polyethylene etc.)
- By LCP, decreasing temperature favours the reaction as it is exothermic, but side reactions will occur.

Catalytic Hydration Process:

C_2H_4 and deionized water. (1 : 0.3 to 1 : 0.8 molar ratio)

250 - 300°C, 6-8 MPa

Ethanol produced is at high temp., so it exchanges heat with feed stream.

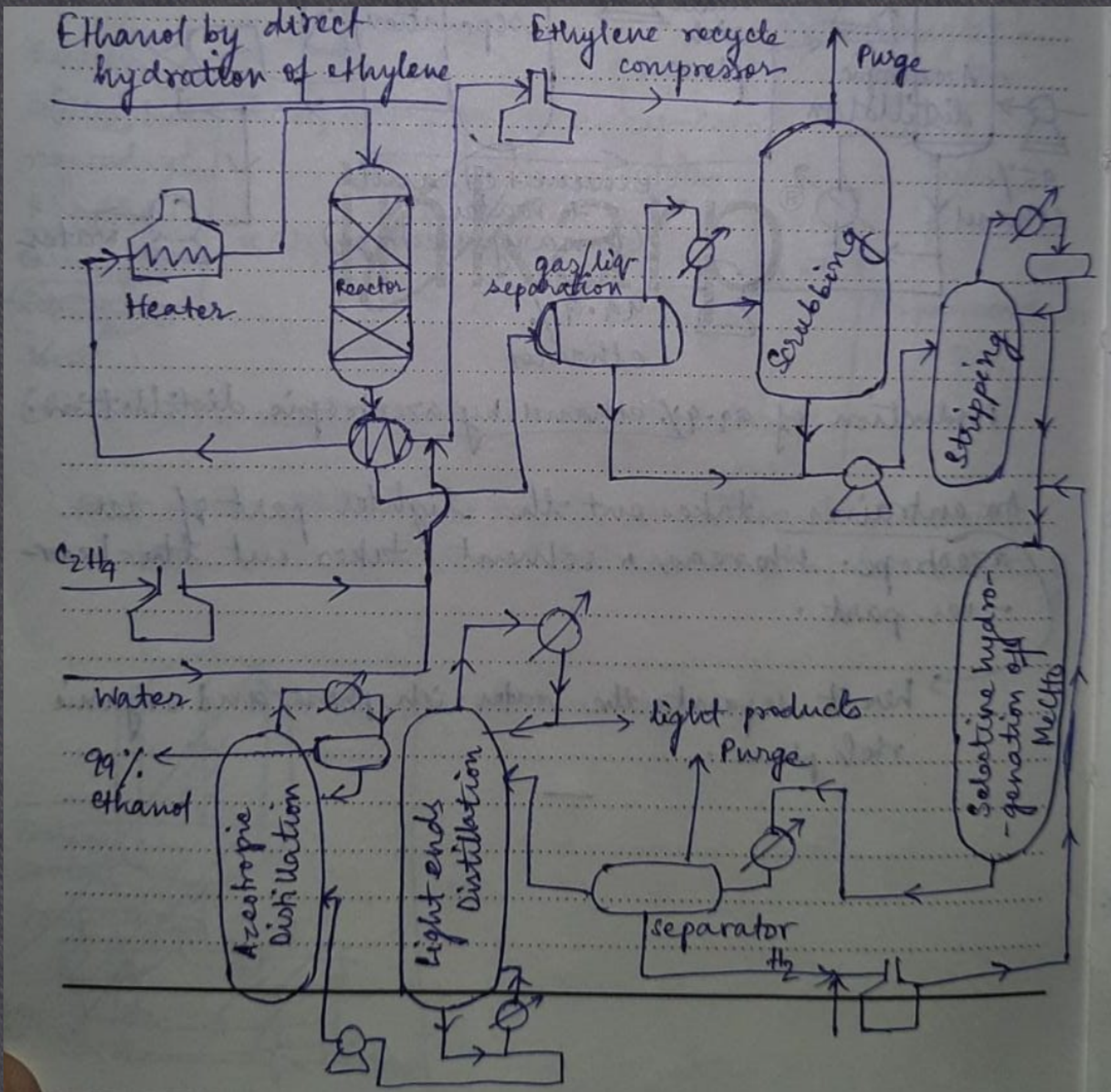
Catalyst : H_3PO_4 on SiO_2.

Feed :- Cooling in Heat Exchanger, followed by water-alcohol cooler. Then sent to distillation column. Bottom product sent to another column, where water-ethanol azeotropic mixture is formed. (95 % ethanol-water).

In azeotropic distillation (~1 atm, very high reflux ratio and NP >> (55 to 65)), upto 99 % ethanol can be obtained.

One column is set in between where product passes through Ni catalyst and H_2, to hydrogenate any possible MeCHO to Et OH.

Ethanol by direct hydration of ethylene

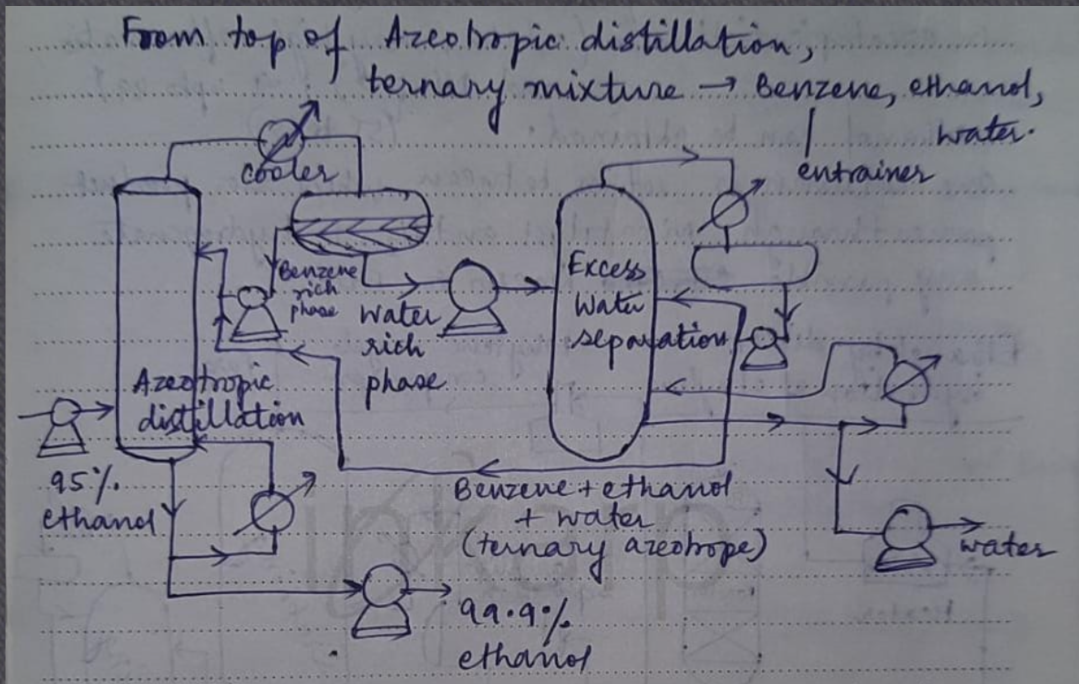


Below 12 kPa pressure, ethanol–water azeotrope breaks down. At 95% ethanol-water, same B.P. in liq., vap. phase is observed. Azeotropic distillation.

From top of Azeotropic distillation,

ternary mixture → Benzene, ethanol, water.

entrainer



Production of 99.9% ethanol by azeotropic distillation.

An entrainer takes out the lighter part of an azeotrope. whereas, a solvent takes out the heavier part.

here, it separates the water rich phase & organic rich phase.

Higher Alcohols:

$C_3 - C_{20}$ are very important industrially. Many compounds can be produced from them.

$$\underbrace{\text{Synthesis gas} + \text{Olefin} \xrightarrow[\text{catalyst}]{\text{Homogenous}} \text{Aldehyde with one C more than olefin}}_{\text{Hydroformylation Reaction (highly exothermic)}}$$

$$\text{Aldehyde} \xrightarrow{\text{Hydrogenation}} \text{Alcohol}$$

Oxo-process

Earlier $Co(CO)_n$, but nowadays Rhodium Carbonyl ($Rh-(CO)_n$) is used, modified by phosphine.

$P < 20$ bar, T (85–115 °C)

$$CO + H_2 + \underset{H}{\overset{R}{>}}C=C\underset{H}{\overset{H}{<}} \xrightarrow{\text{catalyst}} RCH_2CH_2CHO + R-\underset{}{\overset{CH_3}{\overset{|}{C}H}}-CHO$$

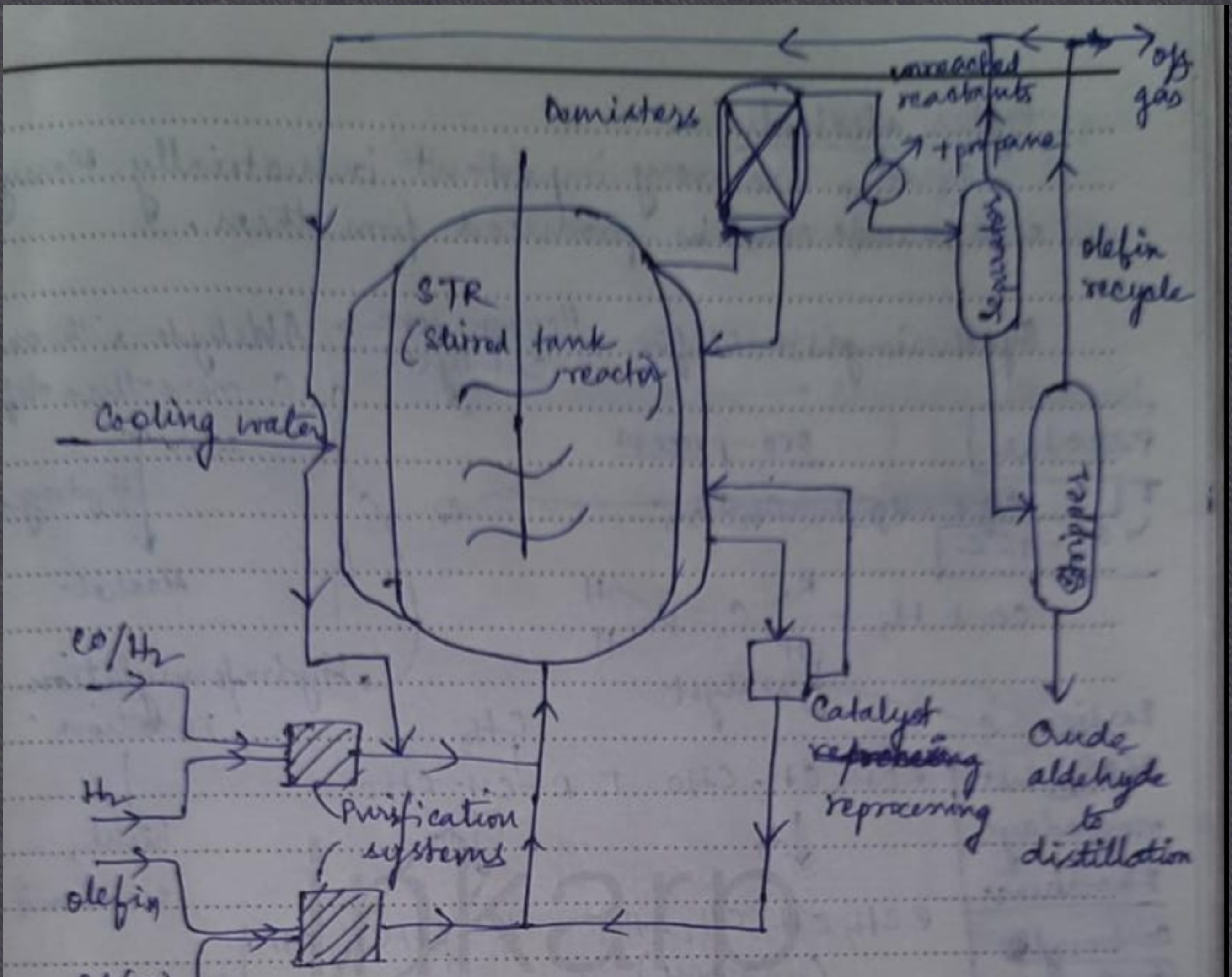
$$RCH_2CH_2CHO \xrightarrow{H_2} RCH_2CH_2CH_2OH$$

$$R-\overset{CH_3}{\overset{|}{C}H}-CHO \xrightarrow{H_2} R-\underset{CH_3}{\underset{|}{C}H}CH_2OH$$

Per pass conversion = 30%.

But $Rh(CO)_n$ is highly sensitive to poisoning.

Problems with $Co(CO)_n$ → (low conversion, low selectivity)



Low pressure oxo process

Oxidation of paraffins to higher alcohols :-

Very unselective and difficult as alkanes are pretty much unreactive. Sometimes aldehydes/ketones are also formed besides alcohols.

Boric Acid & $KMnO_4$ used as catalyst

$\downarrow$ forms borate salts $\uparrow$ hydrolyze

$$\text{Paraffin} + \text{air} \xrightarrow[\substack{0.1 \text{ wt\% } KMnO_4 \\ + 5 \text{ wt\% boric acid}}]{150 - 200^\circ C} \begin{array}{l} 70 \text{ mol\% alcohol} + \\ 20\% \text{ ketones} + \\ 10\% \text{ organic acids} \end{array}$$

Unconverted paraffins obtained by distillation.

Preparation of Ammonia (NH_3) — Haber's Process

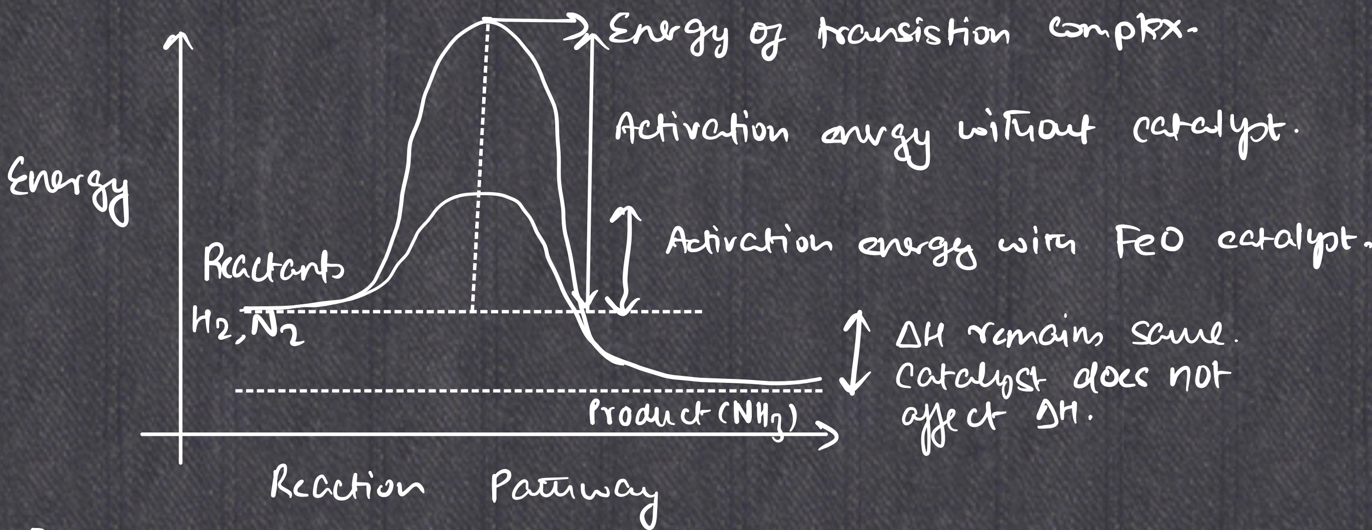
$$\frac{1}{2} N_2 + \frac{3}{2} H_2 \rightleftharpoons NH_3 \quad \text{(exothermic reaction)}$$

Δn_g is -ve, so $P \uparrow \Rightarrow$ more NH_3
$T \downarrow \Rightarrow$ more NH_3

But, optimum temp. is 450 - 500°C
(to break $N \equiv N$ bond and $H-H$)

$P = 150 - 200$ atm

Os, U can produce moderate amount of NH_3, but very costly and rare. FeO is the best catalyst.



[on catalyst surface, adsorbed gases lose their translational degrees of freedom, reducing their energy].

$$K_{p\,eq} = \frac{p_{NH_3}}{p_{N_2}^{1/2}\; p_{H_2}^{3/2}}$$

catalyst → FeO with Al_2O_3, ZrO_2, SiO_2 at about 3% wt + K_2O at about 1%.

Space velocity, space time.

Uses of NH_3 in different industries:

Industry	Use
1.) Fertilizer	For production of $(NH_4)_2SO_4$, $(NH_4)_3PO_4$, NH_4NO_3, NH_2CONH_2
2.) chemicals	Synthesis of HNO_3 (which produces TNT), nitroglycerine, HCN, N_2H_4.

3.) Explosives — NH_4NO_3

4.) Fibre & plastic — Nylon and other polyamides

5.) Refrigeration Industries — Used for making ice, large scale refrigeration plant, air conditioning units.

6.) Pharmaceutical Industries — Drugs such as sulphonamide, anti-malaric drug. Vitamins such as vit-B. (Nicotinamide, thiamine).

7.) Pulp & paper — Ammonium Bisulphate

8.) Mining & Metallurgy — Nitriding (bright annealing) of steel, used in zinc and nickel extraction.

Modern NH_3 plants → one-pass conversion $\sim 15\%$.
$\sim 15,000$ tons/day production.

3 promoters, $P \sim 300$ atm, $T \rightarrow 400$ to $700°C$

New catalyst: Ru/ Graphite → drastically improves the process ($P \downarrow \sim 40$ atm, but same NH_3 production)

<u>Ammonia Production:</u>

Hydrodesulphurization : S-containing compound

(To remove sulphur impurities) + H_2 gas (high P, T)

$H_2S \uparrow \xrightarrow{ZnO} ZnS \downarrow$

$CO, CO_2 \xrightarrow{H_2} CH_4$ (Methanation)

steam reforming and/or shift conversion

Natural Gas (CH_4) feedstock:

1. Sulphur removal from feedstock

$$H_2 + \underset{\text{Mercaptan}}{RSH} \longrightarrow H_2S + RH$$ (R ≡ Alkyl Group)

2. $H_2S + ZnO \longrightarrow H_2O + ZnS \downarrow$ (removal of H_2S)

3. CH_4 steam reforming

$$CH_4 + H_2O \xrightarrow{Fe/Ni} CO + 3H_2$$

4. Shift Conversion

$$CO + H_2O \longrightarrow CO_2 + H_2$$

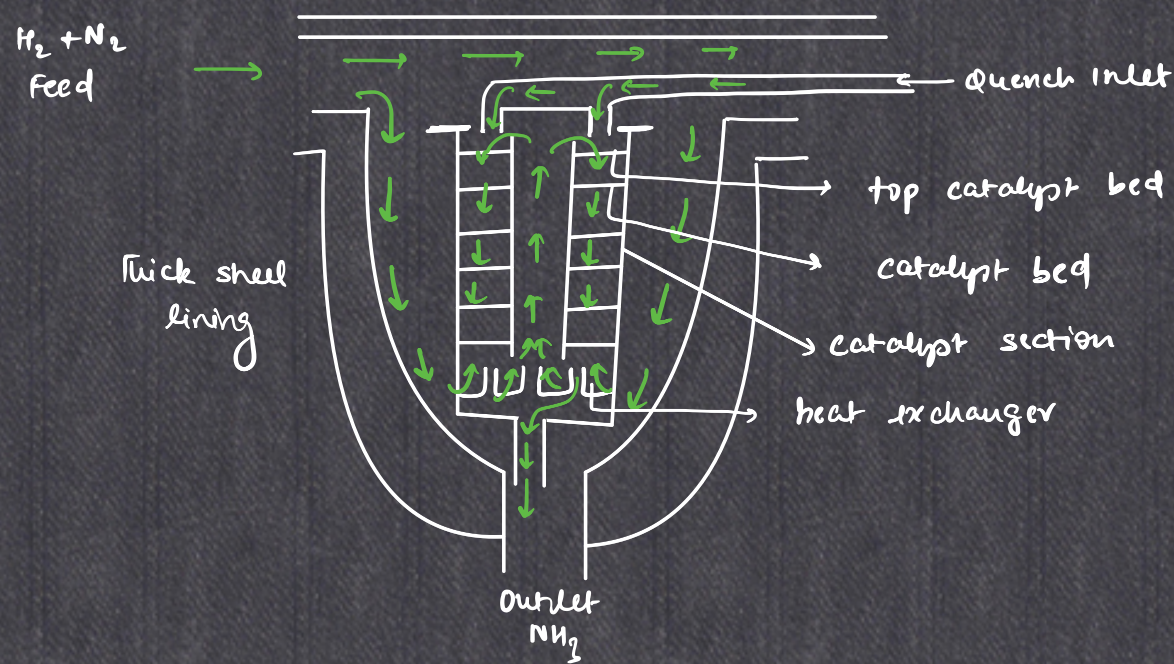
CO_2 is removed either by absorption in ethanolamine or pressure-swing operation.

5. $CO + 3H_2 \longrightarrow CH_4 + H_2O$
$CO_2 + 4H_2 \longrightarrow CH_4 + 2H_2O$
Methanation Reaction (To remove any residual CO/CO_2)

[$N_2 + 3H_2 \underset{60-180 \text{ bar}}{\rightleftharpoons} 2NH_3$ Haber-Bosch Process]

All these processes operate at 25-35 bar.

Ammonia Converter



Heat exchanger exchanges heat between feed and high temp. Ammonia.

Catalyst screens from top to bottom → size increases

Quench Feed (cold) mixes with hot feed (370-425°C) to maintain reaction temp.

↓ to maintain temp.

Unreacted H_2 and N_2 is recycled at each stage.

UREA

(Diamide of carbonic acid)
(Amide of carbamic acid)

$$H_2N - \overset{O}{\overset{\|}{C}} - NH_2$$

$$\underset{gas}{2NH_3 + CO_2} \rightleftharpoons \underset{\text{Ammonium carbamate}}{NH_2COO^- NH_4^+} \quad \text{---------------} \quad (1)$$

$$\underset{solid}{NH_2COO^- NH_4^+} \rightleftharpoons \underset{\text{urea (solid)}}{NH_2CONH_2} + H_2O \quad \text{---------------} \quad (2)$$

① is highly exothermic in nature and ② is endothermic

↓ (large volume change) ↓

150 - 200°C dehydration-decomposition

13.7 - 23.2 MPa (< dehyd. disso. pressure of ②, and > vap. pressure of NH_4COONH_2)

<u>Preparation of solid urea from molten urea</u>

1.) Prilling (a) Fluidized

2.) Granulating (b) Non-fluidized

↓

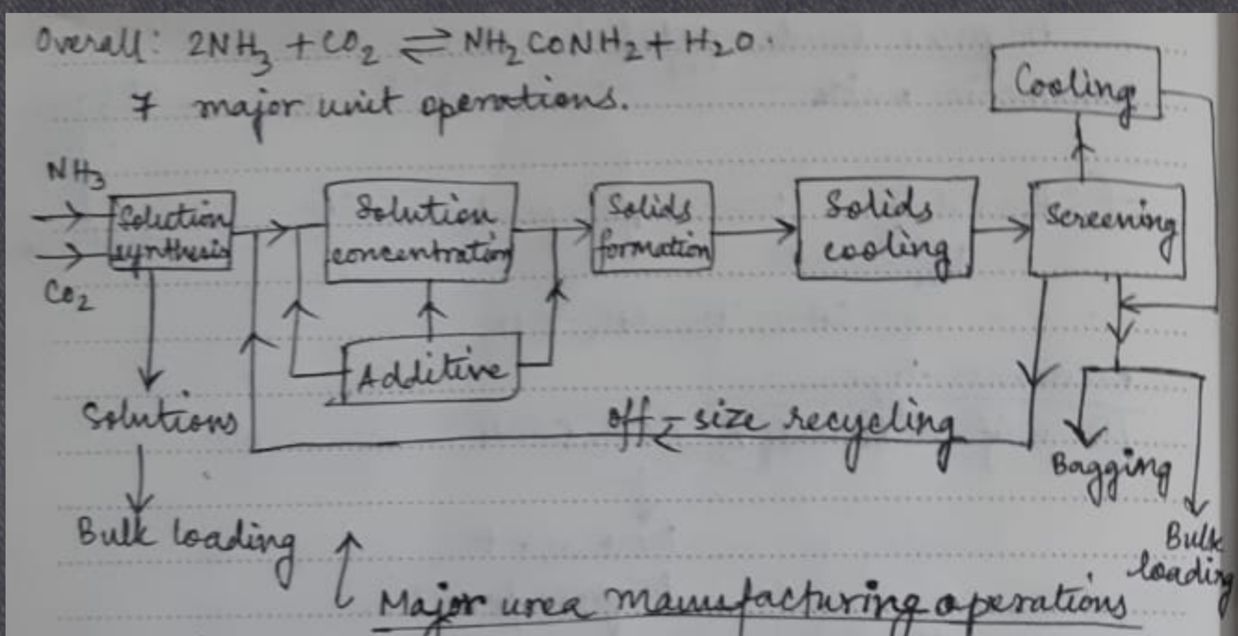
(a) Drum granulation

(b) Pan granulation

Max. conversion (~185°C) is 53% (at $t = \infty$)

Overall: $2NH_3 + CO_2 \rightleftharpoons NH_2CONH_2 + H_2O$

7 major unit operations



Once-through urea process:

Disadvantage ⇒ good amount of NH_3 and undissociated NH_4COONH_2 is lost, and a lot of side products like nitrates, chlorides, sulphates etc.

Solution-recycle process :

$$CO_2, NH_3, NH_4COONH_2 \xrightarrow{H_2O} \text{aq. solution of } NH_4COONH_2$$

↓ recycled to urea production

(makeup raw material)

Urea (DSM) - Stamicarbon BV process

Reactor pressure ~ 130 -150 atm

temp. ~ 170-190 °C

$NH_3 : CO_2$ = 2.8 to 2.9

Stripping unit → P ↓ suddenly.

Water conditioning /water treatment

surface water, ground water

contains dissolved salts of alkali metals and alkaline earth metals. (Ca & Mg bicarbonates, chlorides, sulphates, nitrates, silicates).

hard water, harder than surface water

- $Ca(HCO_3)_2$, $CaCl_2$, $CaSO_4$, $Ca(NO_3)_2$, $CaSiO_3$, $Mg(HCO_3)_2$, $MgCl_2$, $MgSO_4$, $Mg(NO_3)_2$, $MgSiO_3$.

Hardness: Dissolved Ca^{2+} and Mg^{2+} salts as $CaCO_3$ equivalent.

Carbonate
(temporary)
(due to HCO_3^-)

Non-carbonate
(permanent)
(SO_4^{2-}, Cl^-)

TDS : Total Dissolved solids

Units of expressing water analysis : ppm (parts per million)
gpg (grains per gallon)
↳ mg/L

Water conditioning →
- purification (removal of organic matter & micro organisms)
- softening (removing hardness)

Softening

→ cation exchange
→ anion exchange

① Ion Exchange. (Chemical reaction where mobile hydrated ions of exchanger, exchange with hard ions of water, equivalent to equivalent, like charge to charge).

Ion exchange resin/zeolite.

Sodium cation exchanger resin → Na - styrene divinyl benzene sulphonate

Conditions → ~ 150°C
pH → 0 to 14.

Zeolite (Na - alumino silicate)
$NaAlSiO_4$

Resins need regeneration from time to time
(capacity revival)

Anion exchanger resin → Demineralization process
(Mineral ions and salts like SO_4^{2-}, Cl^-, NO_3^-, SiO_3^{2-}, are removed).

Cation exchanger resin

$$Ca \left\{ \begin{matrix} (HCO_3)_2 \\ \ldots \end{matrix} \right. + NaR \longrightarrow \begin{matrix} CaR_2 \\ MgR_2 \end{matrix} + Na_2 \left\{ \begin{matrix} (HCO_3)_2 \\ SO_4 \end{matrix} \right.$$

Mg Cl_2 / SO_4 — salts dissolved in hard water

cation exchanger

insoluble Ca/Mg resins

Cl_2 — sodium salts in water

NaCl sol'n

Regeneration of resin

Anion exchange : Two resin types < Highly basic / weakly basic } For reacting with acidic ions. (Atleast one must be strong)

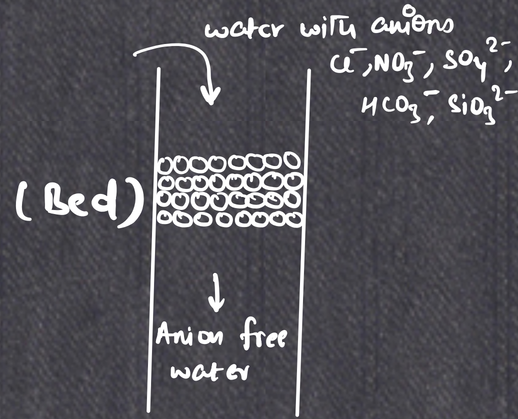
Anion removal reaction by anion exchanger resins

$$H_2SO_4 + 2\,NR_4OH \rightarrow (NR_4)_2SO_4 + 2H_2O$$

(soluble) (insoluble) → Active anion exchanger radical

Highly basic - regenerated by caustic soda

weakly basic - NaOH / NH_4OH



Lime-soda process (Hot, Cold)

↳ (Ca^{2+}, Mg^{2+} removal)

slaked lime, soda ash

$$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3 + 2H_2O$$ (similarly for Mg)

$$MgCO_3 + Ca(OH)_2 \rightarrow Mg(OH)_2 + CaCO_3$$

] CARBONATE HARDNESS

For NON-CARBONATE HARDNESS

① $MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2 + CaCl_2$

② $CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 + 2NaCl$

③ $CaSO_4 + Na_2CO_3 \longrightarrow CaCO_3 + Na_2SO_4$

④ $MgSO_4 + Na_2CO_3 + Ca(OH)_2 \longrightarrow Mg(OH)_2 + CaCO_3 + Na_2SO_4$

Hot lime soda process is operated at B.P. of water. So rate of reaction is high also CO_2 is dissolved and water gets removed.

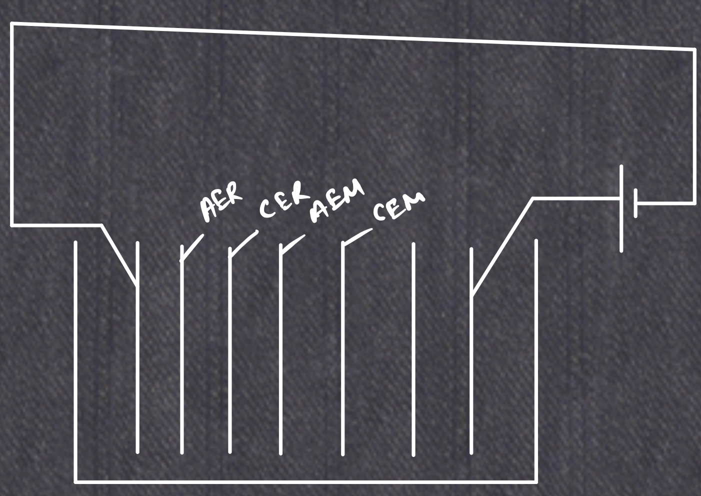
Silica in waste water is removed by dolomite or activated magnesia in the softner.

DEMINERALISATION AND DESALTING

Anion exchange reaction

Reverse osmosis & electrodialysis (you can also achieve some dimineralization

DC current is passed, in the exchanger, cation exchanger takes cations with it and anion exchanger takes anions with it's. There are multiple alternate anion & cation membranes placed next to each other.



RO: Membrane separation takes place, you apply external pressure, higher than osmotic pressure.

MUNICIPAL WASTE WATER TREATMENT

Quality of municipal waste water:

a) Amount of suspended solids.

b) Biochemical Oxygen Demand (BOD)
↳ more BOD ⇒ more sludge

↙

Oxygen required to sustain microbes in water.

→ **PRIMARY TREATMENT:**

① Removal of suspended solids (30-60 %) and reduce B.O.D.

② Screening the large solids.

③ Removal of settalable solids by agglomeration and coagulation.
(LSP)
↳ lime soda

→ **SECONDARY TREATMENT:**

① Oxidation to remove B.O.D to 85-90%.

② Activated sludge system (Activated sludge contains aerobic microorganisms which digest sludge.)

→ **TERTIARY TREATMENT:**

Phosphorous, nitrogen and carbon compounds are removed by chemical treatments

↙

precipitation with lime or metallic hydroxides (e.g. $Al(OH)_3$).